Identifying Key Content and Media Features Driving Online News Popularity

A Comparison of Linear and Non-Linear Models

Objective

The goal of this project is to compare linear and non-linear model to identify which content and media features are most strongly associated with online news popularity.

1. Data Description

The dataset used in this study is the Online News Popularity dataset, which contains information about online news articles and their associated popularity.

In [121…

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import statsmodels.api as sm
```

In [81]:

```
df = pd.read_csv('OnlineNewsPopularity.csv')
df.head()
```

Out[81]:

	url	timedelta	n_tokens_title	n_tokens_content	n_unique_tokens	n_non_stop_words	n_non_stop_unique_tokens	num_hrefs	num_self_hrefs	num_imgs	…	min_positive_p
0	http://mashable.com/2013/01/07/amazon-instant-...	731.0	12.0	219.0	0.663594	1.0	0.815385	4.0	2.0	1.0	…	0.
1	http://mashable.com/2013/01/07/ap-samsung-spon...	731.0	9.0	255.0	0.604743	1.0	0.791946	3.0	1.0	1.0	…	0.
2	http://mashable.com/2013/01/07/apple-40-billio...	731.0	9.0	211.0	0.575130	1.0	0.663866	3.0	1.0	1.0	…	0.
3	http://mashable.com/2013/01/07/astronaut-notre...	731.0	9.0	531.0	0.503788	1.0	0.665635	9.0	0.0	1.0	…	0.
4	http://mashable.com/2013/01/07/att-u-verse-apps/	731.0	13.0	1072.0	0.415646	1.0	0.540890	19.0	19.0	20.0	…	0.

5 rows × 61 columns

In [82]:

```
nrow = df.shape[0]
ncol = df.shape[1]

print(nrow,ncol)
```

```
39644 61
```

In [83]:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 39644 entries, 0 to 39643
Data columns (total 61 columns):
 #   Column                         Non-Null Count  Dtype
---  ------                         --------------  -----
 0   url                            39644 non-null  object
 1    timedelta                     39644 non-null  float64
 2    n_tokens_title                39644 non-null  float64
 3    n_tokens_content              39644 non-null  float64
 4    n_unique_tokens               39644 non-null  float64
 5    n_non_stop_words              39644 non-null  float64
 6    n_non_stop_unique_tokens      39644 non-null  float64
 7    num_hrefs                     39644 non-null  float64
 8    num_self_hrefs                39644 non-null  float64
 9    num_imgs                      39644 non-null  float64
 10   num_videos                    39644 non-null  float64
 11   average_token_length          39644 non-null  float64
 12   num_keywords                  39644 non-null  float64
 13   data_channel_is_lifestyle     39644 non-null  float64
 14   data_channel_is_entertainment 39644 non-null  float64
 15   data_channel_is_bus           39644 non-null  float64
 16   data_channel_is_socmed        39644 non-null  float64
 17   data_channel_is_tech          39644 non-null  float64
 18   data_channel_is_world         39644 non-null  float64
 19   kw_min_min                    39644 non-null  float64
 20   kw_max_min                    39644 non-null  float64
 21   kw_avg_min                    39644 non-null  float64
 22   kw_min_max                    39644 non-null  float64
 23   kw_max_max                    39644 non-null  float64
 24   kw_avg_max                    39644 non-null  float64
 25   kw_min_avg                    39644 non-null  float64
 26   kw_max_avg                    39644 non-null  float64
 27   kw_avg_avg                    39644 non-null  float64
 28   self_reference_min_shares     39644 non-null  float64
 29   self_reference_max_shares     39644 non-null  float64
 30   self_reference_avg_sharess    39644 non-null  float64
 31   weekday_is_monday             39644 non-null  float64
 32   weekday_is_tuesday            39644 non-null  float64
 33   weekday_is_wednesday          39644 non-null  float64
 34   weekday_is_thursday           39644 non-null  float64
 35   weekday_is_friday             39644 non-null  float64
 36   weekday_is_saturday           39644 non-null  float64
 37   weekday_is_sunday             39644 non-null  float64
 38   is_weekend                    39644 non-null  float64
 39   LDA_00                        39644 non-null  float64
 40   LDA_01                        39644 non-null  float64
 41   LDA_02                        39644 non-null  float64
 42   LDA_03                        39644 non-null  float64
 43   LDA_04                        39644 non-null  float64
 44   global_subjectivity           39644 non-null  float64
 45   global_sentiment_polarity     39644 non-null  float64
 46   global_rate_positive_words    39644 non-null  float64
 47   global_rate_negative_words    39644 non-null  float64
 48   rate_positive_words           39644 non-null  float64
 49   rate_negative_words           39644 non-null  float64
 50   avg_positive_polarity         39644 non-null  float64
 51   min_positive_polarity         39644 non-null  float64
 52   max_positive_polarity         39644 non-null  float64
 53   avg_negative_polarity         39644 non-null  float64
 54   min_negative_polarity         39644 non-null  float64
 55   max_negative_polarity         39644 non-null  float64
 56   title_subjectivity            39644 non-null  float64
 57   title_sentiment_polarity      39644 non-null  float64
 58   abs_title_subjectivity        39644 non-null  float64
 59   abs_title_sentiment_polarity  39644 non-null  float64
 60   shares                        39644 non-null  int64
dtypes: float64(59), int64(1), object(1)
memory usage: 18.5+ MB
```

In [84]:

```
df.isna().sum()
```

Out[84]:
```
url                            0
 timedelta                     0
 n_tokens_title                0
 n_tokens_content              0
 n_unique_tokens               0
                              ..
 title_subjectivity            0
 title_sentiment_polarity      0
 abs_title_subjectivity        0
 abs_title_sentiment_polarity  0
 shares                        0
Length: 61, dtype: int64
```

The dataset consists of 39,644 observations and 61 variables. Each observation corresponds to a single news article. Moreover, the dataset includes a mixture of continuous variables and indicator variables capturing temporal, content-related, and metadata attributes of news articles.

2. Y - Exploratory Data Analysis

In [85]:
```
y = df[" shares"]
print(y)
```

```
0         593
1         711
2        1500
3        1200
4         505
         ...
39639    1800
39640    1900
39641    1900
39642    1100
39643    1300
Name:  shares, Length: 39644, dtype: int64
```

In [86]:
```
print(y.describe())
```

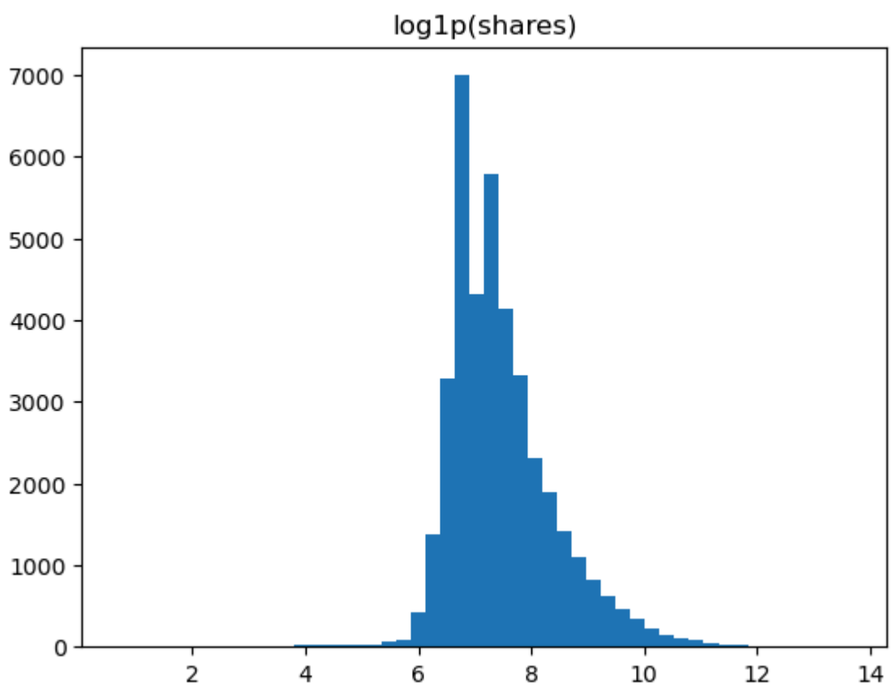
```
count     39644.000000
mean       3395.380184
std       11626.950749
min           1.000000
25%         946.000000
50%        1400.000000
75%        2800.000000
max      843300.000000
Name:  shares, dtype: float64
```

In [87]:
```
print("y.min:", y.min(), "y.max:" , y.max())
```

```
y.min: 1 y.max: 843300
```

Log-Transformed Target Variable

In [88]:
```
plt.hist(np.log1p(y), bins=50)
plt.title("log1p(shares)")
plt.show()
```



Heavy-Tailedness of the Target Variable

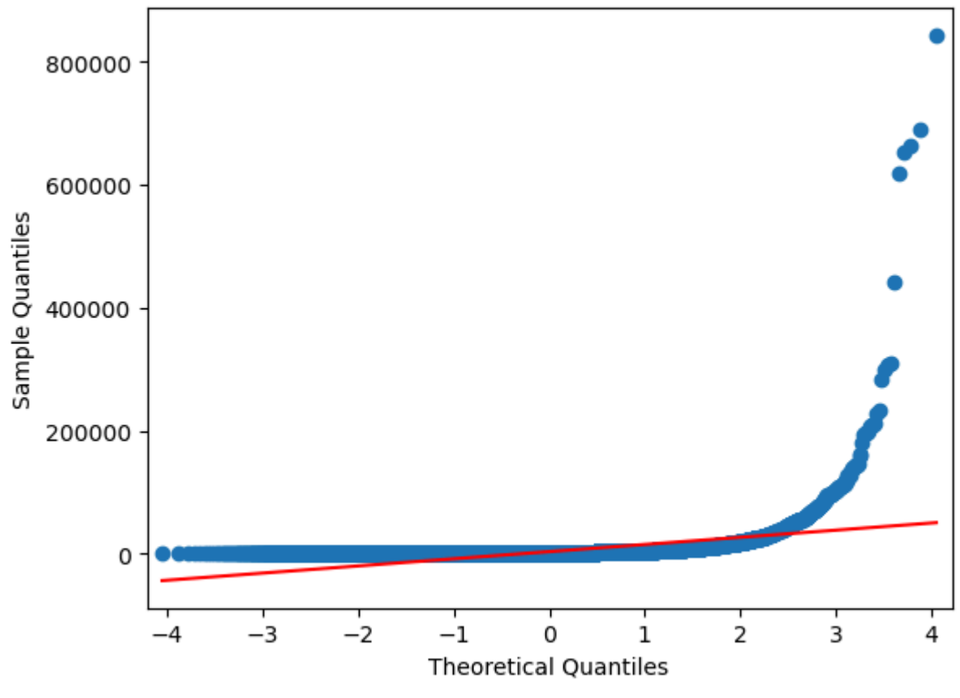
To assess whether the target variable exhibits heavy-tailed behavior, we compare the maximum value to the median as a simple diagnostic. A large max-to-median ratio indicates the presence of extreme values and suggests that the distribution may be heavy-tailed.

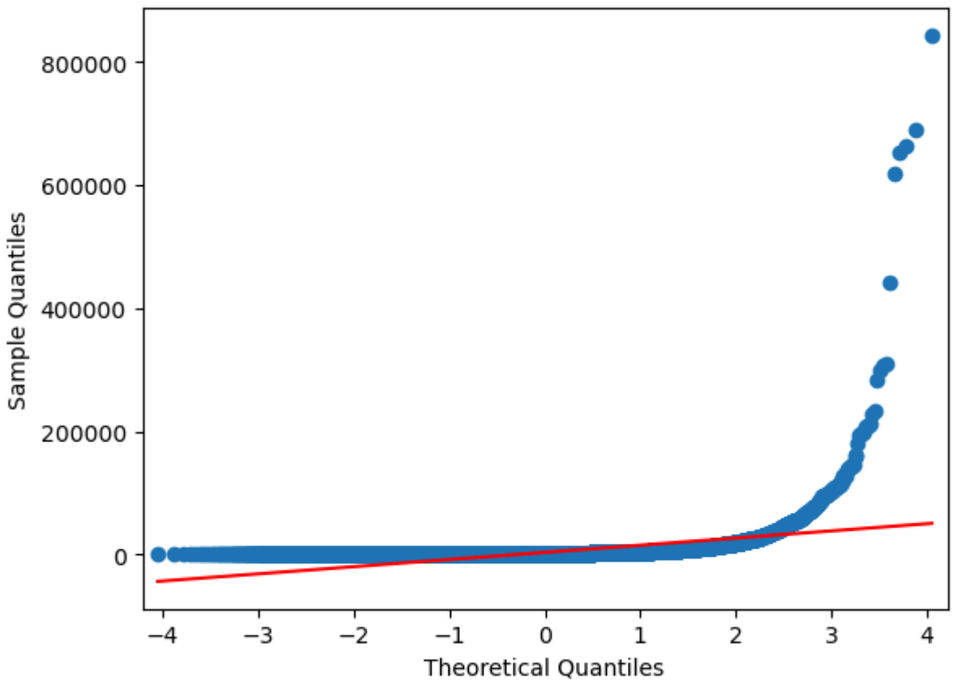
In [89]:
```
print('Max to Med ratio: ' , y.max()/y.median())
```

```
Max to Med ratio:  602.3571428571429
```

In [90]:
```
sm.qqplot(y, line='s')
```

Out[90]:





Extreme values for Target Variable

In [91]:
```
qlow = y.quantile(0.01)
qhigh = y.quantile(0.99)

df[' extreme values'] = ((y <= qlow) | (y >= qhigh))
print(df[' extreme values'].head(10))
```

```
0    False
1    False
2    False
3    False
4    False
5    False
6    False
7    False
8    False
9    False
Name:  extreme values, dtype: bool
```

Based on the max-to-median ratio, the target variable exhibits heavy-tailed behavior. The distribution is strongly right-skewed, with a small number of extreme values dominating the upper tail. This pattern is further supported by the Q–Q plot, which shows substantial deviation from the reference line in the upper quantiles, indicating the presence of numerous extreme observations.

Given the pronounced heavy-tailed nature of the target variable, we adopt Mean Absolute Error (MAE) as the evaluation metric, as it is more robust to extreme values than squared-error-based measures.

3. X - Exploratory Data Analysis

In [92]:
```
X = df.drop(columns=[' shares', 'url'])
```

In [93]:
```
X.head()
```

Out[93]:

	timedelta	n_tokens_title	n_tokens_content	n_unique_tokens	n_non_stop_words	n_non_stop_unique_tokens	num_hrefs	num_self_hrefs	num_imgs	num_videos	...	min_positive_polarity	max_positive_pola
0	731.0	12.0	219.0	0.663594	1.0	0.815385	4.0	2.0	1.0	0.0	...	0.100000	
1	731.0	9.0	255.0	0.604743	1.0	0.791946	3.0	1.0	1.0	0.0	...	0.033333	
2	731.0	9.0	211.0	0.575130	1.0	0.663866	3.0	1.0	1.0	0.0	...	0.100000	
3	731.0	9.0	531.0	0.503788	1.0	0.665635	9.0	0.0	1.0	0.0	...	0.136364	
4	731.0	13.0	1072.0	0.415646	1.0	0.540890	19.0	19.0	20.0	0.0	...	0.033333	

5 rows × 60 columns

In [94]:
```
X.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 39644 entries, 0 to 39643
Data columns (total 60 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0    timedelta                      39644 non-null  float64
 1    n_tokens_title                 39644 non-null  float64
 2    n_tokens_content               39644 non-null  float64
 3    n_unique_tokens                39644 non-null  float64
 4    n_non_stop_words               39644 non-null  float64
 5    n_non_stop_unique_tokens       39644 non-null  float64
 6    num_hrefs                      39644 non-null  float64
 7    num_self_hrefs                 39644 non-null  float64
 8    num_imgs                       39644 non-null  float64
 9    num_videos                     39644 non-null  float64
 10   average_token_length           39644 non-null  float64
 11   num_keywords                   39644 non-null  float64
 12   data_channel_is_lifestyle      39644 non-null  float64
 13   data_channel_is_entertainment  39644 non-null  float64
 14   data_channel_is_bus            39644 non-null  float64
 15   data_channel_is_socmed         39644 non-null  float64
 16   data_channel_is_tech           39644 non-null  float64
 17   data_channel_is_world          39644 non-null  float64
 18   kw_min_min                     39644 non-null  float64
 19   kw_max_min                     39644 non-null  float64
 20   kw_avg_min                     39644 non-null  float64
 21   kw_min_max                     39644 non-null  float64
 22   kw_max_max                     39644 non-null  float64
 23   kw_avg_max                     39644 non-null  float64
 24   kw_min_avg                     39644 non-null  float64
 25   kw_max_avg                     39644 non-null  float64
 26   kw_avg_avg                     39644 non-null  float64
 27   self_reference_min_shares      39644 non-null  float64
 28   self_reference_max_shares      39644 non-null  float64
 29   self_reference_avg_sharess     39644 non-null  float64
 30   weekday_is_monday              39644 non-null  float64
 31   weekday_is_tuesday             39644 non-null  float64
 32   weekday_is_wednesday           39644 non-null  float64
 33   weekday_is_thursday            39644 non-null  float64
 34   weekday_is_friday              39644 non-null  float64
 35   weekday_is_saturday            39644 non-null  float64
 36   weekday_is_sunday              39644 non-null  float64
 37   is_weekend                     39644 non-null  float64
 38   LDA_00                         39644 non-null  float64
 39   LDA_01                         39644 non-null  float64
 40   LDA_02                         39644 non-null  float64
 41   LDA_03                         39644 non-null  float64
 42   LDA_04                         39644 non-null  float64
 43   global_subjectivity            39644 non-null  float64
 44   global_sentiment_polarity      39644 non-null  float64
 45   global_rate_positive_words     39644 non-null  float64
 46   global_rate_negative_words     39644 non-null  float64
 47   rate_positive_words            39644 non-null  float64
 48   rate_negative_words            39644 non-null  float64
 49   avg_positive_polarity          39644 non-null  float64
 50   min_positive_polarity          39644 non-null  float64
 51   max_positive_polarity          39644 non-null  float64
 52   avg_negative_polarity          39644 non-null  float64
 53   min_negative_polarity          39644 non-null  float64
 54   max_negative_polarity          39644 non-null  float64
 55   title_subjectivity             39644 non-null  float64
 56   title_sentiment_polarity       39644 non-null  float64
 57   abs_title_subjectivity         39644 non-null  float64
 58   abs_title_sentiment_polarity   39644 non-null  float64
 59   extreme values                 39644 non-null  bool
dtypes: bool(1), float64(59)
memory usage: 17.9 MB
```

In [95]:
```
X.describe()
```

Out[95]:

	timedelta	n_tokens_title	n_tokens_content	n_unique_tokens	n_non_stop_words	n_non_stop_unique_tokens	num_hrefs	num_self_hrefs	num_imgs	num_videos	...	avg_positive_polarity
count	39644.000000	39644.000000	39644.000000	39644.000000	39644.000000	39644.000000	39644.000000	39644.000000	39644.000000	39644.000000	...	39644.000000
mean	354.530471	10.398749	546.514731	0.548216	0.996469	0.689175	10.883690	3.293638	4.544143	1.249874	...	0.353825
std	214.163767	2.114037	471.107508	3.520708	5.231231	3.264816	11.332017	3.855141	8.309434	4.107855	...	0.104542
min	8.000000	2.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	...	0.000000
25%	164.000000	9.000000	246.000000	0.470870	1.000000	0.625739	4.000000	1.000000	1.000000	0.000000	...	0.306244
50%	339.000000	10.000000	409.000000	0.539226	1.000000	0.690476	8.000000	3.000000	1.000000	0.000000	...	0.358755
75%	542.000000	12.000000	716.000000	0.608696	1.000000	0.754630	14.000000	4.000000	4.000000	1.000000	...	0.411428
max	731.000000	23.000000	8474.000000	701.000000	1.000000	650.000000	304.000000	116.000000	128.000000	91.000000	...	1.000000

8 rows × 59 columns

In [96]:
```
X.nunique().sort_values()
```

Out[96]:
```
extreme values                     2
weekday_is_tuesday                 2
weekday_is_wednesday               2
weekday_is_thursday                2
weekday_is_friday                  2
weekday_is_saturday                2
weekday_is_sunday                  2
is_weekend                         2
data_channel_is_world              2
data_channel_is_tech               2
data_channel_is_socmed             2
weekday_is_monday                  2
data_channel_is_entertainment      2
data_channel_is_bus                2
data_channel_is_lifestyle          2
num_keywords                      10
n_tokens_title                    20
kw_min_min                        26
min_positive_polarity             33
kw_max_max                        35
max_positive_polarity             38
max_negative_polarity             49
num_videos                        53
min_negative_polarity             54
num_self_hrefs                    59
num_imgs                          91
num_hrefs                        133
abs_title_subjectivity           532
abs_title_sentiment_polarity     653
title_subjectivity               673
timedelta                        724
title_sentiment_polarity         813
kw_min_max                      1021
kw_max_min                      1076
self_reference_max_shares       1137
self_reference_min_shares       1255
n_non_stop_words                1451
rate_negative_words             2284
rate_positive_words             2284
n_tokens_content                2406
self_reference_avg_sharess      8626
global_rate_negative_words     10271
global_rate_positive_words     13159
avg_negative_polarity          13841
kw_min_avg                     15982
kw_avg_min                     17003
kw_max_avg                     19438
n_non_stop_unique_tokens       22930
n_unique_tokens                27281
avg_positive_polarity          27301
average_token_length           30136
kw_avg_max                     30834
global_subjectivity            34501
global_sentiment_polarity      34695
LDA_03                         38963
LDA_01                         39098
kw_avg_avg                     39300
LDA_00                         39337
LDA_04                         39370
LDA_02                         39525
dtype: int64
```

Heavy-Tailedness of the Target Variables X

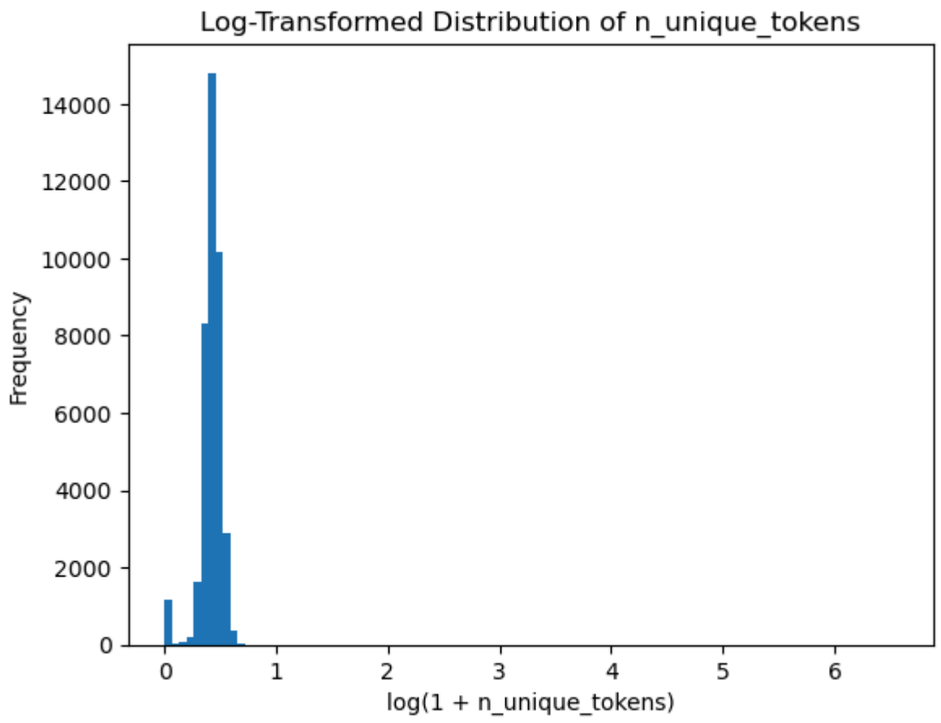
In [97]:
```
X_num = X.select_dtypes(include=['number'])

for col in X_num.columns:
    med = X[col].median()
    if med > 0:
        print(col, X[col].max() / med)
```

```
timedelta 2.1563421828908553
n_tokens_title 2.3
n_tokens_content 20.718826405867972
n_unique_tokens 1300.0125872690435
n_non_stop_words 1042.000004201344
n_non_stop_unique_tokens 941.3793133465615
num_hrefs 38.0
num_self_hrefs 38.666666666666664
num_imgs 128.0
average_token_length 1.7241406215392148
num_keywords 1.4285714285714286
kw_max_min 452.1212121212121
kw_avg_min 181.85926599957537
kw_min_max 602.3571428571429
kw_max_max 1.0
kw_avg_max 3.4480612407111644
kw_min_avg 3.529615208286711
kw_max_avg 68.5081077215125
kw_avg_avg 15.179973274067383
self_reference_min_shares 702.75
self_reference_max_shares 301.17857142857144
self_reference_avg_sharess 383.3181818181818
LDA_00 27.764830810313555
LDA_01 27.768666085496243
LDA_02 22.997714057633765
LDA_03 23.16293015928629
LDA_04 22.765801013315073
global_subjectivity 2.2052790762984444
global_sentiment_polarity 6.1103144196417105
global_rate_positive_words 3.9845397048459485
global_rate_negative_words 12.057534246541778
rate_positive_words 1.4074074074083456
rate_negative_words 3.571428571428571
avg_positive_polarity 2.7874163127185723
min_positive_polarity 10.0
max_positive_polarity 1.25
title_subjectivity 6.666666666666667
abs_title_subjectivity 1.0
```

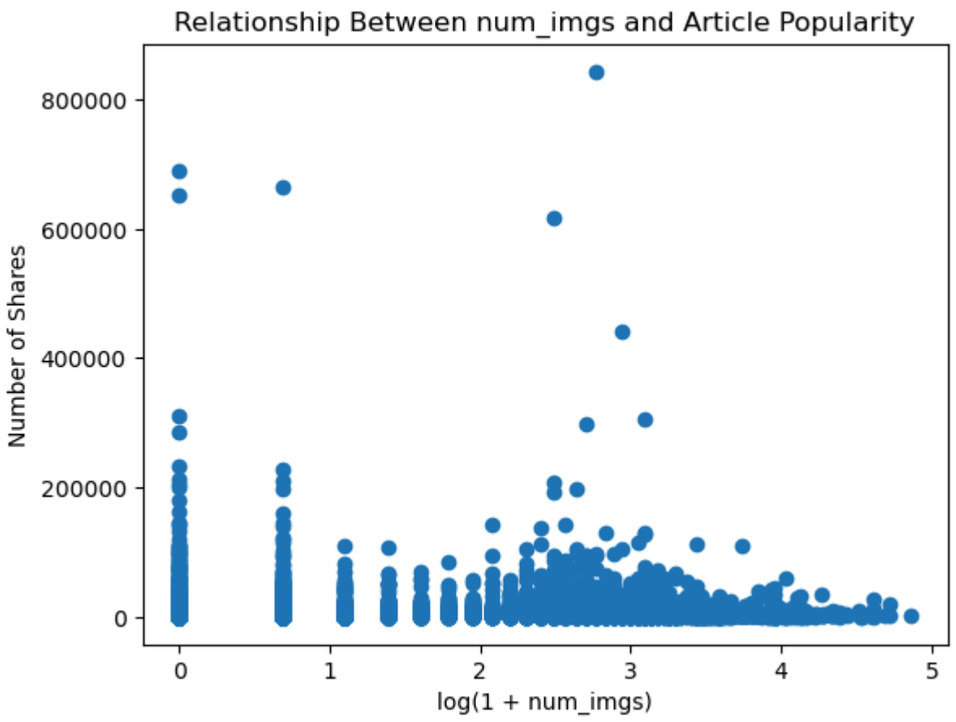
In [98]:
```
plt.hist(np.log1p(X[' n_unique_tokens']), bins=100)
plt.xlabel('log(1 + n_unique_tokens)')
plt.ylabel('Frequency')
plt.title('Log-Transformed Distribution of n_unique_tokens')
```

Out[98]:
```
Text(0.5, 1.0, 'Log-Transformed Distribution of n_unique_tokens')
```



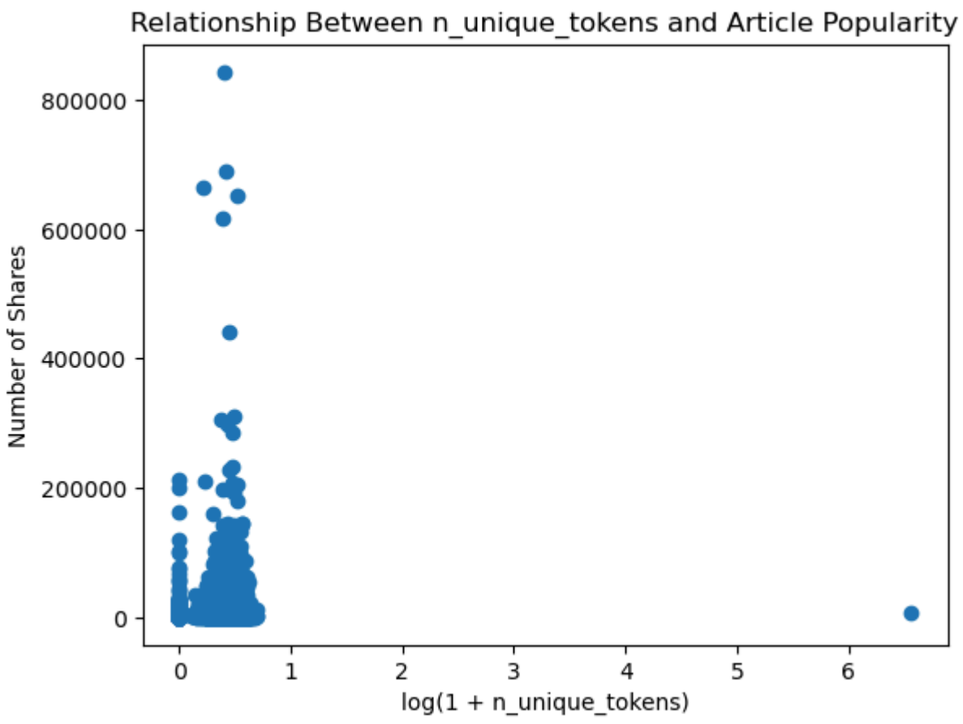
In [99]:
```
plt.scatter(np.log1p(X[' num_imgs']),y)
plt.xlabel('log(1 + num_imgs)')
plt.ylabel('Number of Shares')
plt.title('Relationship Between num_imgs and Article Popularity')
```

Out[99]: Text(0.5, 1.0, 'Relationship Between num_imgs and Article Popularity')



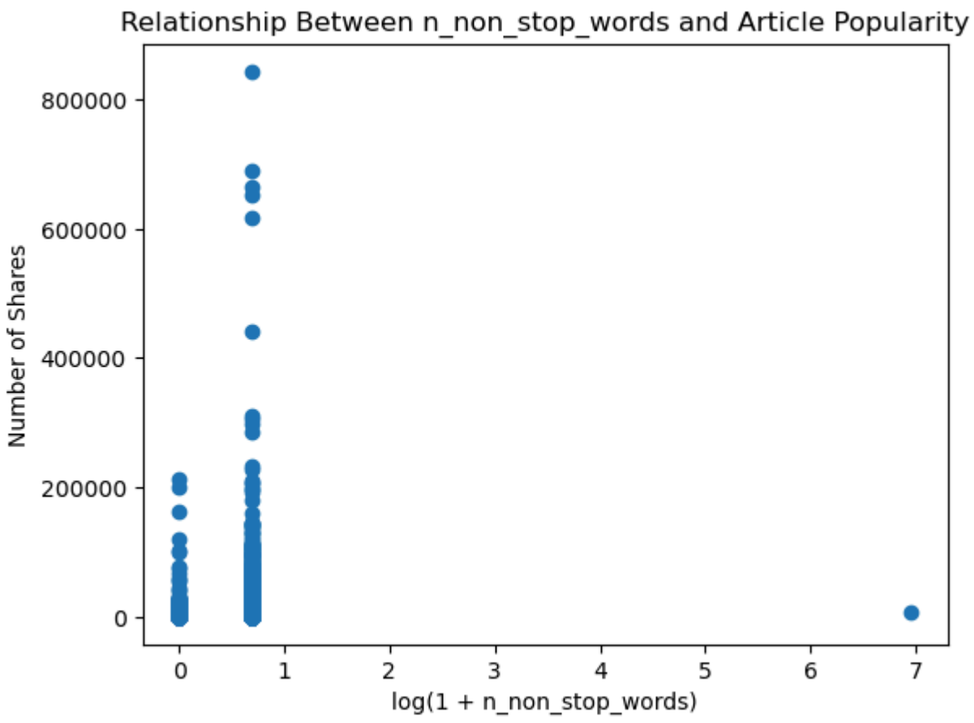
In [100…
```
plt.scatter(np.log1p(X[' n_unique_tokens']),y)
plt.xlabel('log(1 + n_unique_tokens)')
plt.ylabel('Number of Shares')
plt.title('Relationship Between n_unique_tokens and Article Popularity')
```

Out[100… Text(0.5, 1.0, 'Relationship Between n_unique_tokens and Article Popularity')



In [101…
```
plt.scatter(np.log1p(X[' n_non_stop_words']),y)
plt.xlabel('log(1 + n_non_stop_words)')
plt.ylabel('Number of Shares')
plt.title('Relationship Between n_non_stop_words and Article Popularity')
```

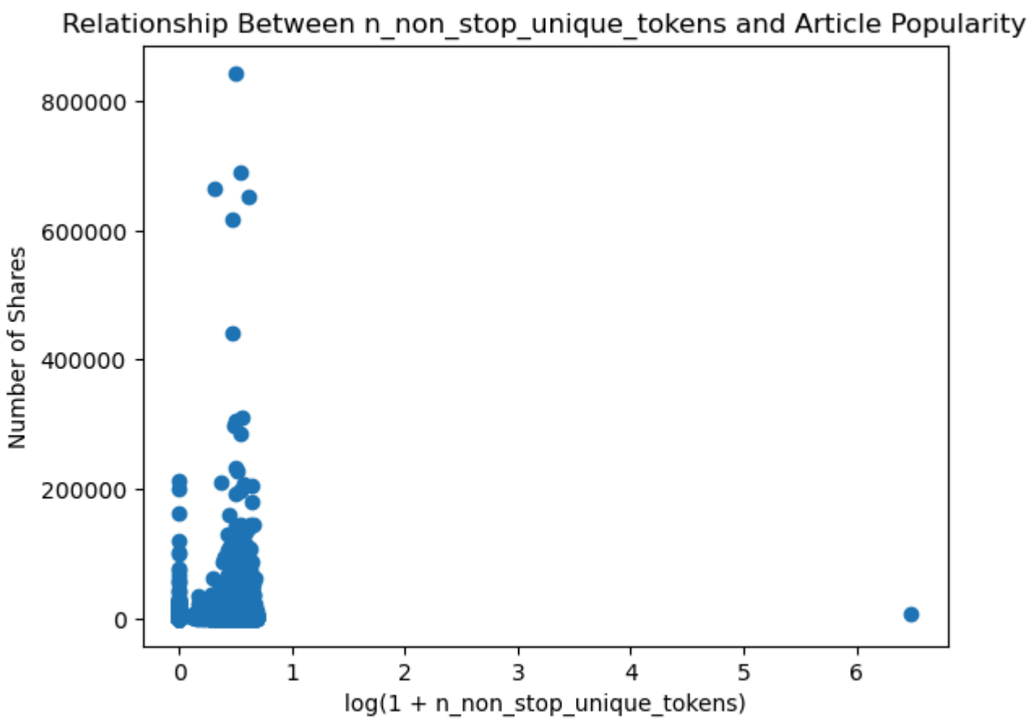
Out[101… Text(0.5, 1.0, 'Relationship Between n_non_stop_words and Article Popularity')



In [102…
```
plt.scatter(np.log1p(X[' n_non_stop_unique_tokens']),y)
plt.xlabel('log(1 + n_non_stop_unique_tokens)')
```

```
plt.ylabel('Number of Shares')
plt.title('Relationship Between n_non_stop_unique_tokens and Article Popularity')
```

Out[102…
```
Text(0.5, 1.0, 'Relationship Between n_non_stop_unique_tokens and Article Popularity')
```



In [103…
```
df.groupby(' is_weekend')[' shares'].mean()
```

Out[103…
```
 is_weekend
0.0    3318.855140
1.0    3903.394412
Name:  shares, dtype: float64
```

Several explanatory variables exhibit extremely large values that may dominate the upper tail of their distributions. To better understand the scale and prevalence of such extreme observations, we visually inspect a subset of variables with particularly long upper tails. The inspection shows that applying a log(1 + x) transformation reduces marginal skewness but does not fully mitigate heavy-tailed behavior or the presence of extreme values. In addition, for indicator variables such as `is_weekend`, we observe that articles published on weekends tend to receive higher numbers of shares.

3. Train–Test Split

In [104…
```
from sklearn.model_selection import train_test_split

X_train,X_test,y_train, y_test = train_test_split(X,y, test_size = 0.33, random_state=42)
```

4. Baseline Model

In [105…
```
from sklearn.metrics import mean_absolute_error
from sklearn.metrics import r2_score

y_pred = y.median()
y_hat = np.repeat(y_pred, len(y_test))

print(f"Mean absolute error {mean_absolute_error(y_test, y_hat):.2f}")
print(f"Coefficient of determination: {r2_score(y_test, y_hat):.2f}")
```

```
Mean absolute error 2511.04
Coefficient of determination: -0.03
```

5. Ordinary Least Squares (OLS)

In [106…
```
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
from sklearn.metrics import mean_absolute_error

regressor = LinearRegression().fit(X_train, y_train)
```

In [107…
```
y_pred1 = regressor.predict(X_test)

print(f"Mean absolute error {mean_absolute_error(y_test, y_pred1):.2f}")

print(f"Coefficient of determination: {r2_score(y_test, y_pred1):.2f}")
```

```
Mean absolute error 2825.47
Coefficient of determination: 0.17
```

Compared to the baseline model, the Ordinary Least Squares (OLS) model achieves a higher MAE, suggesting limited predictive effectiveness. This observation motivates the exploration of potential remedies.

In [108…
```
y_train_log = np.log1p(y_train)

regressor = LinearRegression().fit(X_train, y_train_log)
y_pred2 = regressor.predict(X_test)

y_pred3 = np.expm1(y_pred2)

print(f"Mean absolute error {mean_absolute_error(y_test, y_pred3):.2f}")
print(f"Coefficient of determination: {r2_score(y_test, y_pred3):.2f}")
```

```
Mean absolute error 2447.81
Coefficient of determination: 0.01
```

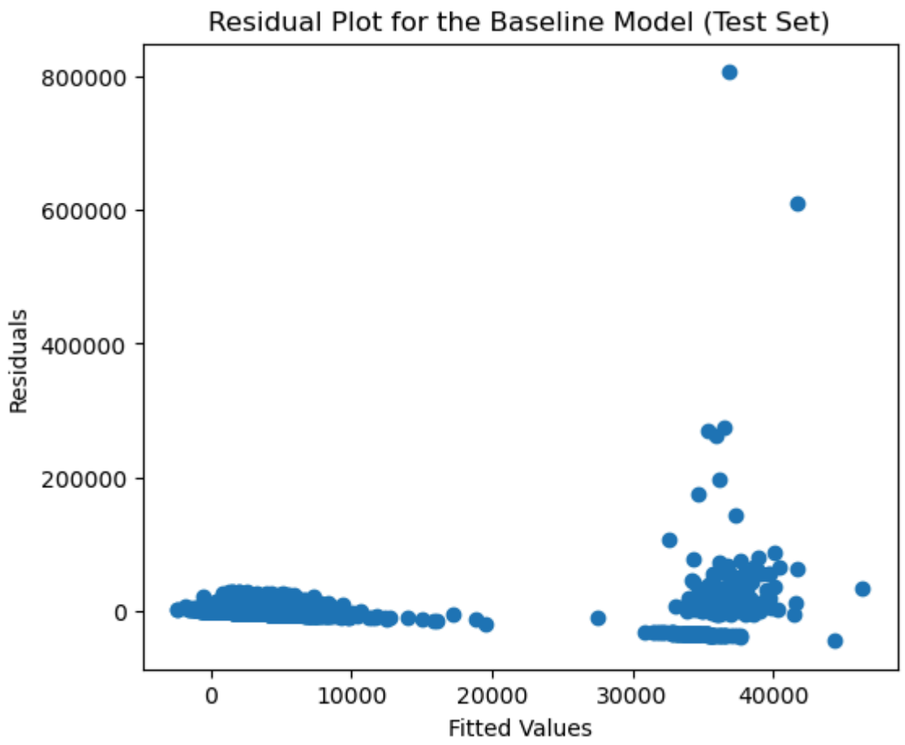
We first apply a log(1 + y) transformation to the target variable in the OLS framework. While this transformation leads to a modest reduction in MAE compared to the original OLS model, it does not sufficiently address the underlying non-linear relationships and extreme-value behavior in the data.

Residual Plot

In [109…
```
residuals = y_test - y_pred1

plt.figure(figsize=(6,5))
plt.scatter(y_pred1, residuals)
plt.xlabel('Fitted Values')
plt.ylabel('Residuals')
plt.title('Residual Plot for the Baseline Model (Test Set)')
```

Out[109…
```
Text(0.5, 1.0, 'Residual Plot for the Baseline Model (Test Set)')
```



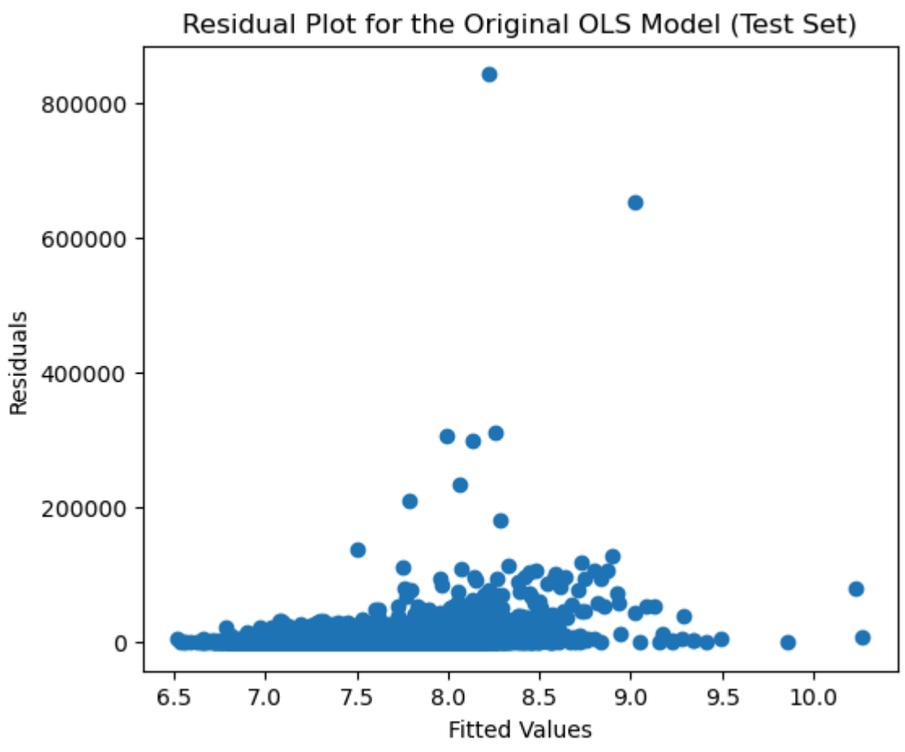
In [110…

```
residuals = y_test - y_pred2

plt.figure(figsize=(6,5))
plt.scatter(y_pred2, residuals)
plt.xlabel('Fitted Values')
plt.ylabel('Residuals')
plt.title('Residual Plot for the Original OLS Model (Test Set)')
```

Out[110…

```
Text(0.5, 1.0, 'Residual Plot for the Original OLS Model (Test Set)')
```



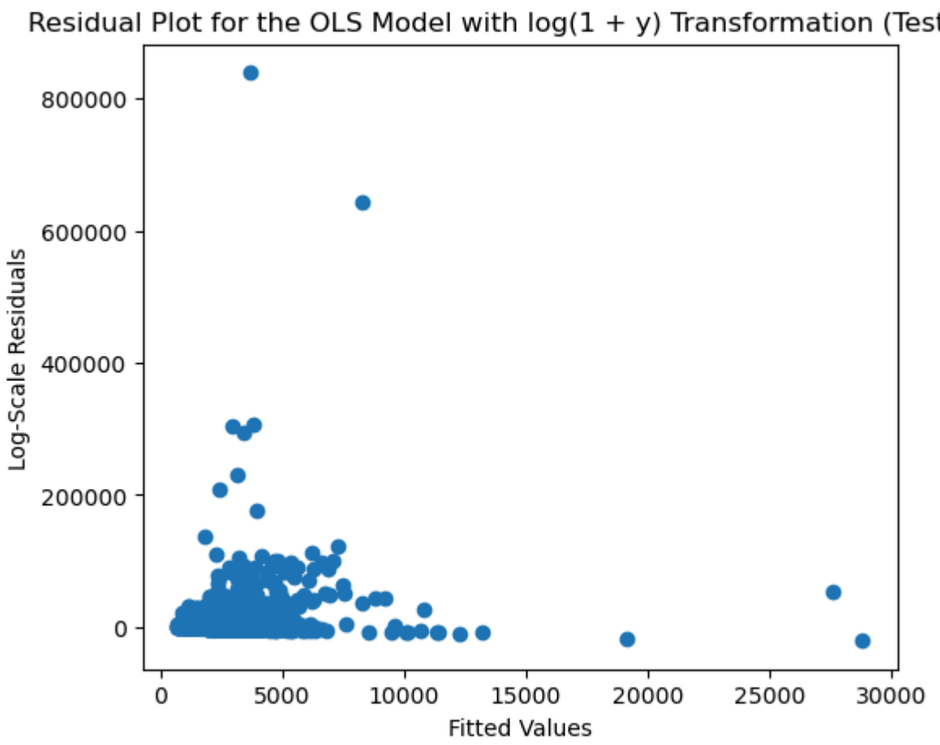
In [111…

```
residuals = y_test - y_pred3

plt.figure(figsize=(6,5))
plt.scatter(y_pred3, residuals)
plt.xlabel('Fitted Values')
plt.ylabel('Log-Scale Residuals')
plt.title('Residual Plot for the OLS Model with log(1 + y) Transformation (Test Set)')
```

Out[111…

```
Text(0.5, 1.0, 'Residual Plot for the OLS Model with log(1 + y) Transformation (Test Set)')
```



To complement the visual inspection of residual plots, we compute the correlation between the absolute residuals and fitted values as a simple diagnostic for heteroskedasticity. A strong positive correlation suggests that residual variance increases with the magnitude of fitted values.

In [112…

```
resid_ols = y_test - y_pred2
corr_ols = np.corrcoef(np.abs(resid_ols), y_pred2)[0,1]

resid_log = y_test - y_pred3
corr_log = np.corrcoef(np.abs(resid_log), y_pred3)[0,1]

print("corr(|resid|, fitted) OLS:", corr_ols)
print("corr(|resid|, fitted) log-OLS:", corr_log)
```

```
corr(|resid|, fitted) OLS: 0.2102246813733072
corr(|resid|, fitted) log-OLS: 0.22978302221950767
```

Comparing the residual plots across the baseline, original OLS, and log-transformed OLS models, clear patterns and clustering of residuals are observed. Even after applying a log(1 + y) transformation, the residuals do not appear to be randomly scattered around zero, indicating persistent model misspecification.

These patterns suggest that the linear model struggles to capture the underlying relationship between the predictors and the target variable, particularly in the presence of noisy features and extreme values. This highlights the limitations of OLS when applied to heavy-tailed and highly heterogeneous data.

6. Random Forest

Given the limitations observed in the linear models, we next consider a non-linear approach using Random Forest to examine whether it provides improved predictive performance.

In [113…

```python
from sklearn.ensemble import RandomForestRegressor

rf = RandomForestRegressor(
    n_estimators = 600,
    max_depth=None,
    min_samples_leaf=2,
    random_state=42,
    n_jobs=-1
)

rf.fit(X_train,y_train)
y_pred_rf = rf.predict(X_test)

print(f"Mean absolute error {mean_absolute_error(y_test, y_pred_rf):.2f}")
print(f"Coefficient of determination: {r2_score(y_test, y_pred_rf):.2f}")
```

```
Mean absolute error 2647.22
Coefficient of determination: 0.24
```

Selection of the Number of Trees

In [114…

```python
from sklearn.ensemble import RandomForestRegressor

for estimator in [100,300,600,800,1000]:
    rf = RandomForestRegressor(
        n_estimators = estimator,
        max_depth=None,
        min_samples_leaf=10,
        random_state=42,
        n_jobs=-1
    )

    rf.fit(X_train,y_train)
    y_pred_rf = rf.predict(X_test)

    print(f"Mean absolute error {mean_absolute_error(y_test, y_pred_rf):.2f}")
    print(f"Coefficient of determination: {r2_score(y_test, y_pred_rf):.2f}")
```

```
Mean absolute error 2553.20
Coefficient of determination: 0.28
Mean absolute error 2552.11
Coefficient of determination: 0.28
Mean absolute error 2551.97
Coefficient of determination: 0.28
Mean absolute error 2551.73
Coefficient of determination: 0.28
Mean absolute error 2553.95
Coefficient of determination: 0.27
```

Selection of Minimum Samples per Leaf

In [115…

```python
from sklearn.ensemble import RandomForestRegressor

for leaf in [1,2,5,10] :
    rf = RandomForestRegressor(
        n_estimators = 600,
        max_depth=None,
        min_samples_leaf=leaf,
        random_state=42,
        n_jobs=-1
    )
    rf.fit(X_train,y_train)
    y_pred_rf = rf.predict(X_test)

    print(f"Mean absolute error {mean_absolute_error(y_test, y_pred_rf):.2f}")
    print(f"Coefficient of determination: {r2_score(y_test, y_pred_rf):.2f}")
```

```
Mean absolute error 2709.13
Coefficient of determination: 0.23
Mean absolute error 2647.22
Coefficient of determination: 0.24
Mean absolute error 2580.25
Coefficient of determination: 0.27
Mean absolute error 2551.97
Coefficient of determination: 0.28
```

In [116…

```python
from sklearn.ensemble import RandomForestRegressor

for leaf in [20,30,40,60,80,100] :
    rf = RandomForestRegressor(
        n_estimators = 600,
        max_depth=None,
        min_samples_leaf=leaf,
        random_state=42,
        n_jobs=-1
    )
    rf.fit(X_train,y_train)
    y_pred_rf = rf.predict(X_test)

    print(f"Mean absolute error {mean_absolute_error(y_test, y_pred_rf):.2f}")
    print(f"Coefficient of determination: {r2_score(y_test, y_pred_rf):.2f}")
```

```
Mean absolute error 2547.34
Coefficient of determination: 0.29
Mean absolute error 2545.78
Coefficient of determination: 0.30
Mean absolute error 2549.60
Coefficient of determination: 0.29
Mean absolute error 2555.91
Coefficient of determination: 0.28
Mean absolute error 2571.00
Coefficient of determination: 0.28
Mean absolute error 2587.56
Coefficient of determination: 0.26
```

Selection of Maximum Features

In [117…

```python
from sklearn.ensemble import RandomForestRegressor

for features in [0.8,0.9,1.0] :
    rf = RandomForestRegressor(
        n_estimators = 600,
        max_depth=None,
        max_features=features,
        min_samples_leaf=30,
        random_state=42,
        n_jobs=-1
    )
    rf.fit(X_train,y_train)
    y_pred_rf = rf.predict(X_test)

    print(f"Mean absolute error {mean_absolute_error(y_test, y_pred_rf):.2f}")
    print(f"Coefficient of determination: {r2_score(y_test, y_pred_rf):.2f}")
```

```
Mean absolute error 2541.07
Coefficient of determination: 0.29
Mean absolute error 2543.71
Coefficient of determination: 0.29
Mean absolute error 2545.78
Coefficient of determination: 0.30
```

Final Model

In [118…

```
from sklearn.ensemble import RandomForestRegressor

rf = RandomForestRegressor(
        n_estimators = 600,
        max_depth=None,
        max_features=0.8,
        min_samples_leaf=30,
        random_state=42,
        n_jobs=-1
    )
rf.fit(X_train,y_train)
y_pred_rf = rf.predict(X_test)

print(f"Mean absolute error {mean_absolute_error(y_test, y_pred_rf):.2f}")
print(f"Coefficient of determination: {r2_score(y_test, y_pred_rf):.2f}")
```

```
Mean absolute error 2541.07
Coefficient of determination: 0.29
```

The final Random Forest configuration is selected based on empirical performance stabilization, balancing predictive performance and computational cost.

Although the Random Forest model does not yield a dramatic reduction in MAE compared to the baseline model, the coefficient of determination (R^2) increases to 0.29. This indicates that the model captures a substantially larger proportion of variance in the target variable, reflecting improved explanatory power.

7. Conclusions

This project compares linear and non-linear modeling approaches to identify factors associated with online news popularity. Using Mean Absolute Error (MAE) as the primary evaluation metric due to the heavy-tailed nature of the target variable, we find that the original Ordinary Least Squares (OLS) model provides limited predictive performance. Diagnostic analyses reveal persistent violations of linear model assumptions, including heteroskedasticity and non-linear relationships, even after applying log transformations and other remedial measures.

To address these limitations, a Random Forest model is employed as a non-linear alternative. While the improvement in MAE relative to the baseline model is modest, the Random Forest achieves an R^2 value of 0.29, indicating a substantial increase in explained variance compared to linear models. Given the highly noisy, heterogeneous, and heavy-tailed nature of the dataset, this level of explanatory power is both reasonable and meaningful.

Overall, the results suggest that non-linear models such as Random Forest are better suited for capturing complex patterns and extreme behavior in online news popularity data, even when gains in absolute prediction error are constrained by inherent data noise.